

$$\begin{aligned}
 &= (p \vee q) \wedge ((\neg p \wedge \neg q) \vee (\neg p \vee \neg q)) \quad (\text{Both De Morgan's Laws}) \\
 &= (p \vee q) \wedge ((\neg p \wedge \neg q) \vee \neg q) \quad (1^{\text{st}} \text{ Associative Law}) \\
 &= (p \vee q) \wedge ((\neg p \vee \neg q) \vee \neg q) \quad (1^{\text{st}} \text{ Commutative Law}) \\
 &= (p \vee q) \wedge (\neg p \vee \neg q) \quad (1^{\text{st}} \text{ Absorption Law})
 \end{aligned}$$

1.9.2 Principal Disjunctive Normal Form

Let p and q be propositional variables. Consider the four conjunctions given below: $p \wedge q, p \wedge \neg q, \neg p \wedge q, \neg p \wedge \neg q$. These four conjunctions are called minterms of p and q .

In general if there are n variables, there will be 2^n minterms. Each minterm is a conjunction in which each variable occurs ^{either} once in the negated form or in the nonnegated form.

Definition 3: For a given formula, an equivalent formula consisting of disjunctions of minterms only is known as its principal disjunctive normal form. Such a normal form is also called sum-of-products canonical form.

Consider the following truth table

p	q	$p \wedge q$	$p \wedge \neg q$	$\neg p \wedge q$	$\neg p \wedge \neg q$
T	T	T	F	F	F
T	F	F	T	F	F
F	T	F	F	T	F
F	F	F	F	F	T

We find that in the columns corresponding to minterms only one entry is T and the other entries are false.

Consider the truth table for $p \rightarrow q$ and $p \leftrightarrow q$

p	q	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

Principal Disjunctive Normal Form of $p \rightarrow q$

$$(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q)$$

Principal Disjunctive Normal Form of $p \leftrightarrow q$

$$(p \wedge q) \vee (\neg p \wedge \neg q)$$